

HARIRAMAKRRISHNAN RAMACHANDRAN

+353 89 970 6156 • hariramakrrish@gmail.com • Dublin, Ireland • Stamp 1G — Full-Time Work Eligible

PROFESSIONAL EXPERIENCE

Software Engineer — HCL Technologies, Chennai, India

Sep 2021 – Jan 2025

- Developed and optimized **3D game simulations** for AI training applications using **Panda3D**, focusing on immersive environments and performance across diverse hardware.
- Designed and implemented **Python codebases** integrated with **Panda3D**, contributing to the core logic of AI training simulations.
- Collaborated with multidisciplinary teams to translate complex requirements into robust technical solutions for simulation development.
- Contributed to **version control workflows** using **GitHub**, managing code branching and pull requests to ensure smooth project collaboration.
- Ensured **cross-platform compatibility** and performance optimization for simulations deployed on various hardware configurations.
- Documented architectural decisions and code implementation details, producing clear technical documentation for collaborative development efforts.
- Participated actively in **code reviews** and team meetings, providing and receiving constructive feedback to enhance code quality and project direction.

SKILLS

Game Development – Panda3D, 3D graphics pipelines, simulation logic, AI training

Programming Languages – Python, C++

Version Control – GitHub, branching, pull requests

Collaboration & Communication – Agile methodologies, code reviews, technical documentation

Operating Systems & Platforms – Cross-platform development, Linux

Additional Tools – Jira, Confluence

PROJECTS

AI Training Simulation Environment (Panda3D)

- Engineered a **Panda3D-based simulation environment** to generate synthetic data for training computer vision models, incorporating realistic physics and environmental conditions.
- Implemented **Python scripts** to procedurally generate complex 3D assets and environments, reducing manual asset creation time by approximately 40%.
- Developed a modular system for integrating different AI training agents, allowing for rapid experimentation with various reinforcement learning algorithms.

Cross-Platform 3D Rendering Engine Integration

- Integrated **Panda3D** with a custom C++ rendering pipeline to achieve enhanced visual fidelity and performance on both Windows and Linux systems.
- Optimized 3D model loading and rendering processes, achieving a **p95 latency reduction** of 180ms for complex scene initialization.
- Authored comprehensive technical documentation for the integration process, enabling easier onboarding for new team members.

EDUCATION

M.Sc. Data Analytics — National College of Ireland, Dublin

Feb 2026

B.E. Computer Science — SNS College of Technology, Coimbatore, India

2021

CERTIFICATIONS

- IELTS Academic: Overall Band Score **7 out of 9**, issued by IELTS Official, Test Date: 28/AUG/2023
- **Full Stack: Java Spring Boot** and Angular (Great Learning)
- Certificate of Completion in Web Development from **University of California**, Davis (Coursera)
- Certificate of Completion in Python programming from **University of Michigan** (Coursera)